2201NSC Linear Algebra and Applications

Week 5

David Harman

Eigenvalues and Eigenvectors

Stable directions, diagonalisation and long-term behaviour

Repeated matrix updates can settle into a pattern

Consider the update

$$\mathbf{x}_{n+1} = A\mathbf{x}_n, \qquad A = \begin{pmatrix} 0.7 & 0.2 \\ 0.3 & 0.8 \end{pmatrix}.$$

Starting from different initial states, repeated multiplication approaches the same proportion:

$$A^{20}\begin{pmatrix} 80 \\ 20 \end{pmatrix} \approx \begin{pmatrix} 40 \\ 60 \end{pmatrix}, \qquad A^{20}\begin{pmatrix} 20 \\ 80 \end{pmatrix} \approx \begin{pmatrix} 40 \\ 60 \end{pmatrix}.$$

The key observation

The limiting direction satisfies

$$A\begin{pmatrix} 40 \\ 60 \end{pmatrix} = \begin{pmatrix} 40 \\ 60 \end{pmatrix}.$$

It is unchanged by the transformation.

Eigenvectors are directions that a matrix does not turn

For most vectors, $A\mathbf{v}$ points in a new direction. An eigenvector is different: the output remains on the same line.

$$\boxed{A\mathbf{v} = \lambda\mathbf{v}}$$

$\lambda > 1$	$0 < \lambda < 1$	$\lambda < 0$
Stretch along the eigenvector direction.	Shrink along the eigenvector direction.	Reverse direction and scale by $\lvert\lambda\rvert$.

The eigenvector must be nonzero

Let A be an $n \times n$ matrix.

Definition

A scalar λ is an **eigenvalue** of A if there is a **nonzero** vector $\mathbf{v}$ such that

$$A\mathbf{v} = \lambda\mathbf{v}.$$

The vector $\mathbf{v}$ is an eigenvector corresponding to λ.

Why exclude $\mathbf{0}$?

Every matrix satisfies $A\mathbf{0} = \lambda\mathbf{0}$ for every scalar λ. Including the zero vector would make the definition meaningless.

If $\mathbf{v}$ is an eigenvector, every nonzero multiple $c\mathbf{v}$ is also an eigenvector for the same eigenvalue.

A nonzero solution requires a singular matrix

Starting from the eigenvalue equation,

$$\begin{aligned} A\mathbf{v} &= \lambda\mathbf{v}, \\ A\mathbf{v} - \lambda I\mathbf{v} &= \mathbf{0}, \\ (A - \lambda I)\mathbf{v} &= \mathbf{0}. \end{aligned}$$

For this homogeneous system to have a nonzero solution, $A - \lambda I$ must be singular.

Therefore,

$$\boxed{\det(A - \lambda I) = 0.}$$

This equation is the **characteristic equation**; its left-hand side is the **characteristic polynomial**.

Use a three-step method

1. Write the characteristic equation
$$\det(A - \lambda I) = 0.$$
2. Solve it to find the eigenvalues $\lambda_1, \lambda_2, \ldots$.
3. For each eigenvalue λ_i, solve
$$(A - \lambda_i I)\mathbf{v} = \mathbf{0}$$
and choose any convenient nonzero vector from the solution family.

Keep the roles separate

The determinant finds *eigenvalues*; the homogeneous system finds the corresponding *eigenvectors*.

Whiteboard practice — find the eigenpairs

For each matrix, find all eigenvalues and a corresponding eigenvector for each eigenvalue.

1

$$A = \begin{pmatrix} 3 & 2 \\ 4 & 1 \end{pmatrix}$$

2

$$B = \begin{pmatrix} 6 & -4 \\ 3 & -1 \end{pmatrix}$$

3

$$C = \begin{pmatrix} 3 & -1 \\ 1 & 1 \end{pmatrix}$$

4

$$D = \begin{pmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}$$

Check: Does the sum of the eigenvalues equal the trace, and does their product equal the determinant?

Whiteboard check — eigenpairs

Eigenvectors are shown using one convenient representative; any nonzero multiple is equivalent.

1

$$\lambda = 5 : \quad (1,1)^{\mathsf{T}}$$
$$\lambda = -1 : \quad (1,-2)^{\mathsf{T}}$$

2

$$\lambda = 2 : \quad (1,1)^{\mathsf{T}}$$
$$\lambda = 3 : \quad (4,3)^{\mathsf{T}}$$

3

$$\lambda = 2 \text{ (repeated)} : \quad (1,1)^{\mathsf{T}}$$

Only one independent eigenvector direction.

4

$$\lambda = 2 \text{ (repeated)} : \quad (1,0,0)^{\mathsf{T}},\ (0,1,0)^{\mathsf{T}}$$
$$\lambda = 3 : \quad (1,0,1)^{\mathsf{T}}$$

There are two independent eigenvectors for $\lambda = 2$.

Trace and determinant provide fast checks

If an $n \times n$ matrix has eigenvalues $\lambda_1, \ldots, \lambda_n$, counting multiplicity, then

$$\boxed{\lambda_1 + \cdots + \lambda_n = \operatorname{tr}(A)}$$

$$\boxed{\lambda_1 \lambda_2 \cdots \lambda_n = \det(A).}$$

For a 2×2 matrix, the characteristic equation can be written as

$$\boxed{\lambda^2 - \operatorname{tr}(A)\lambda + \det(A) = 0.}$$

Use these as checks

They can detect an arithmetic error, but they do not identify the eigenvectors.

Triangular matrices reveal their eigenvalues immediately

For any upper- or lower-triangular matrix, the eigenvalues are the diagonal entries, counting repetitions.

$$A = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 2 & 1 \\ 0 & 0 & 1 \end{pmatrix} \qquad \Longrightarrow \qquad \lambda = 1,\ 2,\ 1.$$

Why?

$A - \lambda I$ is still triangular, so its determinant is the product of its diagonal entries:

$$\det(A - \lambda I) = (1-\lambda)^2(2-\lambda).$$

The eigenvectors must still be found by solving $(A - \lambda I)\mathbf{v} = \mathbf{0}$.

Real matrices can have complex eigenvalues

Consider the 90° rotation matrix

$$R = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

Its characteristic equation is

$$\det(R - \lambda I) = \lambda^2 + 1 = 0,$$

so

$$\lambda = \pm i.$$

Geometric meaning

A 90° rotation has no nonzero real direction that remains on its original line. Complex eigenvalues occur as conjugate pairs for real matrices.

A repeated eigenvalue may not supply enough directions

The matrix

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

has the repeated eigenvalue $\lambda = 1$, but

$$(A - I)\mathbf{v} = \mathbf{0} \qquad \Longrightarrow \qquad v_2 = 0.$$

Thus all eigenvectors lie along the single direction

$$\mathbf{v} = c \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad c \neq 0.$$

Important distinction

An eigenvalue can be repeated algebraically without producing the same number of linearly independent eigenvectors.

A basis gives unique coordinates for every vector

A set $\{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ is a **basis** for $\mathbb{R}^n$ when it is linearly independent and spans $\mathbb{R}^n$. Every vector then has a unique representation

$$\mathbf{x} = c_1\mathbf{v}_1 + \cdots + c_n\mathbf{v}_n.$$

Writing the basis vectors as columns,

$$V = \begin{pmatrix} | & & | \\ \mathbf{v}_1 & \cdots & \mathbf{v}_n \\ | & & | \end{pmatrix}, \qquad \mathbf{x} = V\mathbf{c}.$$

Coordinates in the new basis

The coordinate vector is $\mathbf{c} = V^{-1}\mathbf{x}$ whenever V is invertible.

Use a determinant to test n vectors in $\mathbb{R}^n$

Any basis for $\mathbb{R}^n$ contains **exactly n vectors**: fewer cannot span $\mathbb{R}^n$, while more cannot be linearly independent.

Place the vectors into the columns of a square matrix

$$V = \begin{pmatrix} | & & | \\ \mathbf{v}_1 & \cdots & \mathbf{v}_n \\ | & & | \end{pmatrix}.$$

Then

$$\boxed{\det(V) \neq 0 \quad \Longleftrightarrow \quad \mathbf{v}_1, \ldots, \mathbf{v}_n \text{ are linearly independent.}}$$

$\det(V) \neq 0$

V is invertible, so its columns form a basis for $\mathbb{R}^n$.

$\det(V) = 0$

At least one column depends on the others; they do not form a basis.

Distinct eigenvalues produce independent eigenvectors

Key theorem

Eigenvectors corresponding to distinct eigenvalues are linearly independent.

Therefore, if an $n \times n$ matrix has n distinct eigenvalues, its eigenvectors form a basis for $\mathbb{R}^n$ (or $\mathbb{C}^n$ when complex eigenvalues are used).

Why this matters

Every vector can be written as

$$\mathbf{x} = c_1\mathbf{v}_1 + \cdots + c_n\mathbf{v}_n,$$

so the effect of A can be analysed one eigenvector direction at a time.

Eigenvectors make the matrix diagonal

Suppose A has n linearly independent eigenvectors. Put them in the columns of V, and put the matching eigenvalues on the diagonal of D:

$$V = \begin{pmatrix} | & & | \\ \mathbf{v}_1 & \cdots & \mathbf{v}_n \\ | & & | \end{pmatrix}, \qquad D = \mathrm{diag}(\lambda_1, \ldots, \lambda_n).$$

Because $A\mathbf{v}_i = \lambda_i \mathbf{v}_i$,

$$AV = VD.$$

Multiplying by V^{-1} gives

$$\boxed{A = VDV^{-1}.}$$

This factorisation is a **diagonalisation** of A.

Match each column of V with its entry in D

Suppose

$$A\mathbf{v}_1 = 2\mathbf{v}_1, \qquad A\mathbf{v}_2 = -3\mathbf{v}_2.$$

Then a consistent arrangement is

$$V = \begin{pmatrix} | & | \\ \mathbf{v}_1 & \mathbf{v}_2 \\ | & | \end{pmatrix}, \qquad D = \begin{pmatrix} 2 & 0 \\ 0 & -3 \end{pmatrix}.$$

Swapping the columns of V is allowed only if the diagonal entries of D are swapped as well.

A matrix is diagonalizable when

It has enough linearly independent eigenvectors to make V invertible. Distinct eigenvalues guarantee this; repeated eigenvalues require a check.

Diagonalisation turns matrix powers into scalar powers

If $A = VDV^{-1}$, then

$$A^2 = (VDV^{-1})(VDV^{-1}) = VD^2V^{-1},$$
$$A^n = VD^nV^{-1}.$$

For

$$D = \mathrm{diag}(\lambda_1, \ldots, \lambda_n),$$

we simply have

$$D^n = \mathrm{diag}(\lambda_1^n, \ldots, \lambda_n^n).$$

The payoff

The difficult operation A^n becomes a set of ordinary scalar powers.

Diagonalisation also simplifies inverses

If $A = VDV^{-1}$ and every eigenvalue is nonzero, then

$$A^{-1} = VD^{-1}V^{-1},$$

where

$$D^{-1} = \operatorname{diag}\left(\frac{1}{\lambda_1}, \ldots, \frac{1}{\lambda_n}\right).$$

All $\lambda_i \neq 0$

D and A are invertible.

Some $\lambda_i = 0$

D and A are singular.

This agrees with $\det(A) = \lambda_1 \cdots \lambda_n$.

Eigenvalues control long-term behaviour

Write an initial state in an eigenvector basis:

$$\mathbf{x}_0 = c_1\mathbf{v}_1 + \cdots + c_n\mathbf{v}_n.$$

After k updates,

$$A^k\mathbf{x}_0 = c_1\lambda_1^k\mathbf{v}_1 + \cdots + c_n\lambda_n^k\mathbf{v}_n.$$

- Terms with $|\lambda_i| < 1$ decay.
- Terms with $|\lambda_i| > 1$ grow.
- Terms with $\lambda_i < 0$ alternate direction.
- The eigenvalue of largest magnitude usually dominates.

A steady state is an eigenvector with eigenvalue 1

For an update $\mathbf{x}_{n+1} = A\mathbf{x}_n$, a steady state satisfies

$$A\mathbf{x} = \mathbf{x}.$$

Therefore,

$$\boxed{A\mathbf{x} = 1\mathbf{x}.}$$

Procedure

1. Find an eigenvector associated with $\lambda = 1$.
2. Scale it to match the interpretation of the state.

For a probability distribution, scale the entries so that they sum to 1, not so that the vector has Euclidean length 1.

The dominant eigenvector explains stable proportions

In population and transition models, the total size may change while the relative proportions approach a fixed pattern.

$$A^k \mathbf{x}_0 = c_1 \lambda_1^k \mathbf{v}_1 + \sum_{i=2}^{n} c_i \lambda_i^k \mathbf{v}_i.$$

If

$$|\lambda_1| > |\lambda_i| \quad \text{for } i \geq 2,$$

then the other components become small relative to the first.

Interpretation

$\mathbf{v}_1$ gives the long-run proportions; λ_1 gives the long-run growth or decay factor per update.

Link-based rankings are eigenvector problems

Assign each page a score proportional to the scores of pages linking to it. With a suitable link matrix C, the ranking rule has the form

$$C\mathbf{r} = \lambda\mathbf{r}.$$

Interpretation

The dominant positive eigenvector $\mathbf{r}$ supplies relative page scores. Normalising its entries makes the ranking easier to compare.

This same principle appears in ranking competitors, webpages and nodes in a network: importance is inherited from important neighbours.

Short Revision Quiz

Eigenpairs, bases, diagonalisation and applications

Question 1 — Definition

Which equation defines an eigenpair $(\lambda, \mathbf{v})$ for A?

A. $A\mathbf{v} = \lambda\mathbf{v}$ with $\mathbf{v} \neq \mathbf{0}$

B. $A\mathbf{v} = \lambda A$

C. $A + \mathbf{v} = \lambda\mathbf{v}$

Question 2 — Characteristic equation

Which equation is solved to find the eigenvalues of A?

A. $\det(A) = \lambda$

B. $\det(A - \lambda I) = 0$

C. $(A - \lambda I)\mathbf{v} = \mathbf{1}$

Question 3 — Verify an eigenpair

Let

$$A = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}, \qquad \mathbf{v} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

Is $\mathbf{v}$ an eigenvector of A? If so, what is its eigenvalue?

A. Yes, $\lambda = 2$

B. Yes, $\lambda = 3$

C. No

Question 4 — Trace and determinant

A 2×2 matrix has eigenvalues 4 and -2. What are its trace and determinant?

A. $\operatorname{tr}(A) = 2$, $\det(A) = -8$

B. $\operatorname{tr}(A) = -8$, $\det(A) = 2$

C. $\operatorname{tr}(A) = 6$, $\det(A) = 8$

Question 5 — Triangular matrix

What are the eigenvalues of

$$A = \begin{pmatrix} 5 & 2 & -1 \\ 0 & -3 & 4 \\ 0 & 0 & 5 \end{pmatrix}?$$

A. $5, -3, 5$

B. $2, -3, 4$

C. $5, -3, -1$

Question 6 — Diagonalisation

In $A = VDV^{-1}$, what forms the columns of V?

A. The rows of A

B. Linearly independent eigenvectors of A

C. The eigenvalues of A

Question 7 — Matrix powers

If $A = VDV^{-1}$, which expression equals A^n?

A. $V^n D^n V^{-n}$

B. $VD^n V^{-1}$

C. $V^{-1} D^n V$

Question 8 — Long-term behaviour

After n repeated updates, an eigenvector component is scaled by λ^n.
Which component decays to zero as the number of updates $n \to \infty$?

A. $\lambda = 1.2$

B. $\lambda = -1$

C. $\lambda = 0.4$

Question 9 — Steady states

A nonzero steady state of $\mathbf{x}_{n+1} = A\mathbf{x}_n$ is an eigenvector associated with which eigenvalue?

A. 0

B. 1

C. The determinant of A

Question 10 — Repeated eigenvalues

If a 3×3 matrix has a repeated eigenvalue, which statement is always true?

A. The matrix cannot be diagonalised

B. The matrix is automatically diagonal

C. We must check whether there are three independent eigenvectors

Quiz answers 1–5

1 — A

$A\mathbf{v} = \lambda\mathbf{v}$ with $\mathbf{v} \neq \mathbf{0}$.

2 — B

Eigenvalues satisfy $\det(A - \lambda I) = 0$.

3 — B

$A\mathbf{v} = \begin{pmatrix} 3 \\ 3 \end{pmatrix} = 3\mathbf{v}$.

4 — A

The trace is $4 + (-2) = 2$ and the determinant is $(4)(-2) = -8$.

5 — A

The eigenvalues of a triangular matrix are its diagonal entries: $5, -3, 5$.

Quiz answers 6–10

6 — B

The columns of V are linearly independent eigenvectors of A.

7 — B

$A^n = VD^nV^{-1}$ because the middle factors $V^{-1}V$ cancel.

8 — C

$0.4^n \to 0$ because $|0.4| < 1$.

9 — B

A steady state satisfies $A\mathbf{x} = \mathbf{x} = 1\mathbf{x}$.

10 — C

A repeated eigenvalue may or may not yield enough independent eigenvectors; the eigenspace must be checked.

Final takeaway

Eigenvectors reveal stable directions; diagonalisation uses those directions to simplify powers and explain long-term behaviour.